Chapter 13, Problem 75P

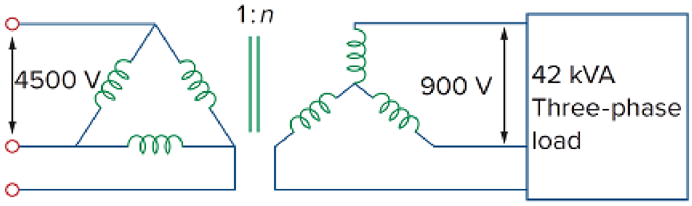
Problem

A balanced three-phase transformer bank with the Δ-Y connection depicted in Fig. 13.137 is used to step down line voltages from 4,500 V rms to 900 V rms. If the transformer feeds a 120-kVA load, find:

(a) the turns ratio for the transformer,

(b) the line currents at the primary and secondary sides.

Figure 13.137



Step-by-step solution

Step 1 of 3

(a)

Refer to Figure 13.137, in the textbook.

The line voltage on the primary side is V_{Lp} .

The line voltage on the secondary side is V_{Ls} .

The relation between V_{Lp} and V_{Ls}, for a $\Delta - \mathrm{Y}$ connection is,

$$V_{Ls} = n\sqrt{3}V_{Lp}$$

The turns ratio of the $\Delta - \mathrm{Y}$ connected transformer is $1 : n$.

The value of n is,

$$n = \frac{V_{Ls}}{\sqrt{3}V_{Lp}}$$

Substitute 4500 V rms for V_{Lp} and 900 V rms for V_{Ls}.

$$\begin{aligned} n &= \frac{900}{\sqrt{3} \times 4500} \\ &= 0.1154 \end{aligned}$$

The value of n is 0.1154.

Thus, the turns ratio is $\boxed{1 : 0.1154}$.

Step 2 of 3

(b)

The line current at the secondary side is,

$$S_T = \sqrt{3}V_{Ls}I_{Ls}$$

Substitute 900 V rms for V_{Ls} and $120\ \text{KVA}$ for S_T .

$$120000 = \sqrt{3}\times 900 I_{Ls}$$
$$I_{Ls} = \frac{120000}{900\sqrt{3}}$$

$$120000 = \sqrt{3}\times 900 I_{Ls}$$
$$I_{Ls} = 76.9$$

Thus, the line current on the secondary side of the transformer is $\boxed{76.9\ \text{A}}$.

Step 3 of 3

The relation between the primary line current and secondary line current in a $\Delta - \text{Y}$ connected transformer is,

$$I_{Ls} = \frac{I_{Lp}}{n\sqrt{3}}$$

Substitute 76.9 A for I_{Ls} and 0.1154 for n.

$$76.9 = \frac{I_{Lp}}{0.1154\sqrt{3}}$$
$$I_{Lp} = 15.37\ \text{A}$$

Thus, the line current on the primary side of the transformer is $\boxed{15.37\ \text{A}}$.